



Reading: Summary and Highlights

Congratulations! You have completed this module. At this point, you know that:

Image captioning with Meta's Llama

- Combining computer vision with natural language processing creates powerful tools for understanding visual content.
- Three main stages of the image captioning process with a multimodal large language model (LLM) are:
 - Input processing
 - Image validation and encoding
 - Multimodal LLM processing
- Input processing receives and prepares the image and optional text prompt.
- Image validation and encoding validate and convert the image into a format (e.g., Base64) suitable for the model.
- Multimodal LLM processing combines visual and textual information to generate a descriptive caption.
- Core components of the image captioning system to produce captions tailored to prompts are:
 - Visual encoders
 - Text embedding
 - Fusion layers
 - Language generation tools
- Implementing an image captioning system using Meta's Llama 4 Maverick model via IBM watsonx involves:
 - Importing libraries and authenticating access
 - Encoding images and preparing prompts
 - Sending combined image-text messages to the model
 - Extracting descriptive text from the model's response

Text-to-video generation with OpenAI's Sora